The Witness-Weave Cosmos

F4 · Final Wildcard Candidate · Explicitly Fictional, Stipulated, and Dismantled

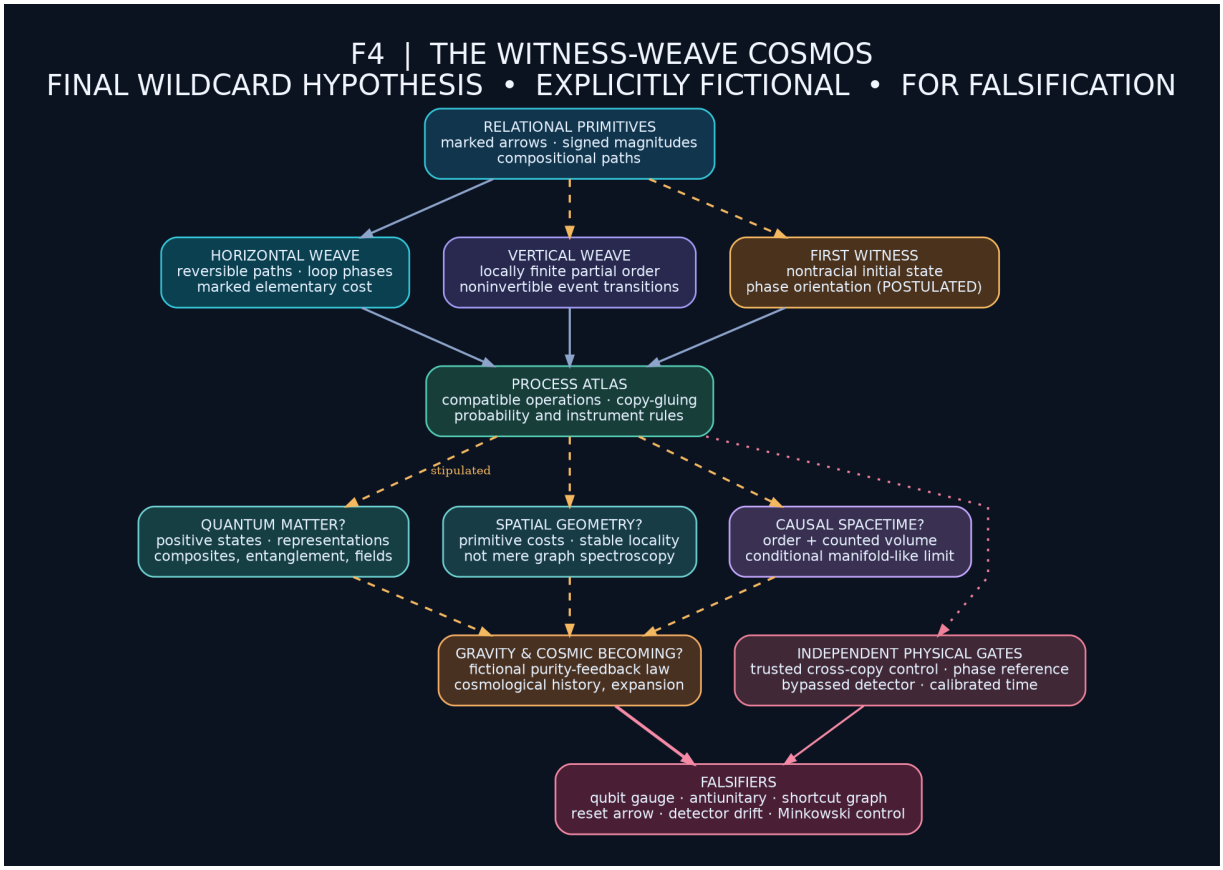
Session C · From Prephysical Hypotheses to Physical Models

7 November 2026 UTC



An auditable cosmological architecture

The **horizontal** and **vertical** branches are different stipulations; dashed amber lines are not proved physical bridges. The trusted experimental gate is independent input, not a consequence of prephysical structure.



Research program: From Prephysical Hypotheses to Physical Models
Research date: 7 November 2026 UTC (6 November, Vancouver)
Status: Fictional, non-empirical cosmology. A stipulated model and a finite mathematical test suite, **not** a derived physical theory.
Disposition: Final wildcard episode, after which strict scientific research resumes without automatic cosmological stipulations.

> **Ontological thesis, deliberately extreme:** Before spacetime there is a weave of composable distinctions. A world is an internally coherent set of ways of making and registering distinctions. Reversible closed relations carry gauge-like phase; irreversible transitions supply a candidate causal order; coherent representations produce candidate matter; stable order plus volume becomes candidate geometry. The universe is not situated in spacetime. Spacetime is one late, coarse-grained appearance of the weave.

This is a **proposal for demolition**, not a discovery or an interpretation of measured cosmology. Every arrow toward real physics below requires an independent bridge.

I. Source audit before invention

1. **Process identification (Session D).** Two minimal, physically admissible qubit models have the same every allowed single-copy adaptive sequence probability and different named preparation purities, 5/8 and 25/32. The similarity is linear but not a reversible transformation of the full qubit state space. A separately calibrated entangling gate separates the named preparations with same-sign outcome probabilities 41/64 and 19/32. Independent finite-matrix reconstruction reproduces these exact fractions. The full-scale crossed-detector Bayes crossings are 45 and 39 trials at 10% mean risk for two specified instruments; the sharp coupling-2 ROC upper limit at p1=1/2 is 0.848167975587…. These are synthetic and conditional instrument calculations.
2. **Marked relations (Sessions B/C).** Preserving signed edge generators and cycle holonomies avoids the information loss in magnitude-only L, but the generated algebra M_n(C) alone forgets which operations were primitive. Entirely reversible path reachability does not define an antisymmetric causal order. Completely positive channels T rho T* forget scalar phases unless an independently calibrated coherent path comparison is available.
3. **Operational atlas.** An inverse limit of finitely compatible observational descriptions can introduce points not in the original model class. All finite compatible descriptions need not imply a globally realizable *physical* universe. The meaning of ‘equivalent’ depends on the operational instrument and the allowed gauge, including antiunitary coconjugation.
4. **Foundational baseline.** The October 30/31 Wilson-Dirac and 2D compact-group results remain separate mathematical workstreams. The stronger fixed-positive-mass Wilson finite-part expression has substantive separate manuscript proofs and imported operator-domain, concentration, boundary, and cross-boundary dependencies. None of this is proof of the cosmology proposed here.

Independent checks are implemented in `verify_f4.py` and `verified_results.json`. They verify the qubit purity and CNOT probabilities, one-copy gauge cancellation, 45/39 full-record crossings, detector ROC values, and the fictional feedback inconsistency. They do not verify actual apparatuses, original source-archive manifests, or all mathematical claims from the previous sessions.

II. The formal F4 system

The **Witness-Weave** is a family

$$\mathbb{W} = (P, \prec, \{\mathcal{G}_p\}, \{\Phi_{pq}\}, \mathcal{M}, \{\omega_p\}, \mathcal{D}, \mathbf{w}, \chi),$$

where P is an event set, prec is a relation of precedence, $\mathcal{G}_p$ are locally marked path categories of reversible relational moves, Φ_{pq} are composition-respecting transitions, $\mathcal{M}$ is a

stipulated parallel-composition rule with cross-system operations, ω_p are candidate positive states, $\mathcal{D}$ is a positive decoherence functional or operational probability structure, **w** are coupling weights, and χ are holonomy phases. None is presumed to have physical meaning merely from its definition.

The ten F4 axioms (fictional, explicit)

W1. Relational primacy. There are no initially supplied coordinates, physical particles or clock values. Primitive data are labeled composable relations and their admissible compositions. The existence and *choice* of primitive relations are stipulated.

W2. The horizontal weave. Each proto-event p has a connected marked path category $\mathcal{G}_p$ with elementary arrows e, formal reverses and a loop-phase map $\chi_p(\gamma) \in U(1)$. Phase composition obeys $\chi(\gamma\delta) = \chi(\gamma)\chi(\delta)$. The elementary marks cannot be inferred from the bare generated operator algebra.

W3. The vertical weave. The event relation (P,prec) is locally finite and antisymmetric. The world is not declared to have derived this order from reversible arrows. F4 boldly stipulates that the vertical structure is an intrinsic primitive, and proposes to explain its physical significance afterward.

W4. Compatible growth. For p prec q prec r there are process maps with $\Phi_{qr} \circ \Phi_{pq} = \Phi_{pr}$. A history is an admissible compositional path through these maps. A consistent atlas comprises nested finite descriptions with a specified mathematical completion. Realizability in the original model class is an additional condition.

W5. Quantum witness. The full world admits a positive, normalized history-decoherence functional $\mathcal{D}$, a coherent complex probability calculus, and a faithful monoidal rule for combining independent systems. F4 stipulates these ingredients; it does not derive Born probability, purification, CAR, CCR or all quantum correlations from graph path composition.

W6. First Witness. A nontracial boundary state ω_* and an oriented reference phase are selected at a primitive cosmic boundary. This 'first witness' is the fictional asymmetry that allegedly seeds matter, chirality and the experienced temporal direction. It is an added boundary postulate, not a theorem from symmetric data.

W7. Locality via elementary cost. A candidate spatial separation is the minimal cost of a sequence of marked elementary reversible operations: $d_p(i,j) = \inf_{\gamma: i \to j} \sum_{e \in \gamma} c_p(e)$. The costs and the generating family are part of the model. Mere generated matrix algebras do not fix d.

W8. Lorentzian realization. Where the vertical partial order is manifold-like, and a separately normalized event density measures volume, the candidate Lorentzian metric is recovered from causal order plus volume. Most finite partial orders are not thereby assumed manifold-like. The continuum Hawking–King–McCarthy–Malament result has nontrivial hypotheses.

W9. Matter and gravity feedback. Candidate particles are persistent representation sectors of the marked path/process structure. F4's deliberately strong *proposed* local feedback rule is

$$\boxed{\log[w_e(q)/w_e(p)] = -\gamma\,(\,\mathrm{Tr}\,\rho_e(q)^2 - \mathrm{Tr}\,\rho_e(p)^2),\ \gamma > 0.}$$

It claims a local change in state purity physically changes relational coupling and hence geometry. This is a concrete falsification target. It is **not** Einstein's equation and has no experimentally derived gamma or metric correspondence.

W10. Cosmic becoming. In a coarse-grained branch with event density n and an inferred antichain volume V, F4 stipulates a candidate scale factor a proportional to $V^{1/3}$ and an irreversible growth process. Local phase defects allegedly form fields, graph/order excitations form matter, coherent historical branching yields effective quantum probabilities, and evolving couplings curve the spacetime reconstruction. No Friedmann equation, Lorentz covariance, 3+1 dimensions, Standard Model spectrum, entropy arrow, cosmological redshift

relation or unit conversion has been derived.

A microscopic sum-over-histories rule

For any *finite cutoff* N, F4 assigns amplitudes to admissible histories h,

$$A_N(h) = Z_N^{-1/2} \exp\{i[\alpha S_{\text{order}}(h) + \beta S_{\text{hol}}(h) + S_{\text{matter}}(h)]/\hbar_*\},$$

with $S_{\text{hol}}(h) = \sum_{\ell \in \text{loops}(h)}[1 - \text{Re}\chi(\ell)]$. It assigns positive history kernel $D_N(h, h') = A_N(h)\overline{A_N(h')}$. For a family of decoherent coarse-grained alternatives $\{C_a\}$, stipulate $P(C_a) = \sum_{h,h' \in C_a} D_N(h, h')$ with appropriate normalization. This constructs a nonnegative rank-one kernel at finite N but **does not** guarantee decoherence or additivity for arbitrary alternatives or existence of a normalized continuum theory. Choice of action, complex amplitudes, $\hbar_*$, boundary conditions and physical apparatus are explicit imports.

III. The fictional chronology of a universe

The Unwitnessed: An ensemble of composable signed distinctions without physical coordinate values. Multiple loop histories remain possible.

The First Witness: An explicitly selected nontracial boundary condition, breaking the formal symmetry between some descriptions. This does not prove there was an actual unique first event.

The Causal Loom: Locally finite ordered transitions begin to organize reversible local relations into a candidate past/future. If its order statistics become manifold-like and volume units are externally calibrated, an effective Lorentzian geometry becomes conceivable.

Knots and matter: Persistent holonomy and representation sectors become the model's *candidate* particles and field configurations. They have no demonstrated correspondence to electron spin, chirality, mass or the Standard Model.

The bending weave: The hypothetical purity-feedback law changes local weights and costs. In the story it is experienced as gravity, but its actual predictions do not match any established law without further derivation.

The expanding tapestry: New causally compatible transitions accumulate and effective spatial sections may grow. The connection to measured redshift, physical expansion and dark energy remains open.

The narrative is the creative steelman, not the evidential conclusion.

IV. Exact mathematical footholds (not fundamental physics)

1. Marked graph paths recover hop distance **relative to chosen elementary generators**, and loops retain signed/complex holonomies when their marked amplitudes are retained.
2. A sufficiently rich operational atlas can determine a *minimal real linear predictive representation up to similarity*, rather than uniquely determine every quantum state and effect.
3. For a finite predictive representation $(s, \ell, \{T_a\})$, any invertible S gives $s' = Ss$, $\ell' = \ell S^{-1}$, $T'_a = ST_aS^{-1}$, and every word probability is unchanged. For n independent copies and only product operations, the same holds with $S^{\otimes n}$. An independently marked nonproduct operation may break the equivalence.
4. When a spacetime is future/past distinguishing, causal order determines conformal metric under the classical continuum hypotheses, with volume fixing the remaining factor. A generic finite poset need not approximate such spacetime.
5. A two-coupling detector ROC and a crossed-source reference generate genuine conditional falsification tests; passing them does not independently establish the required detector stationarity or reference anchoring.

V. Deliberate demolition of F4

D1. The quantum postulate is imported. W5 states positive decoherence functional and monoidal composition. A noncommutative relation algebra alone does not derive Born's rule or a positive normalized family of instruments. Contextuality alone permits nonsignalling postquantum models.

D2. No operationally unique state. Two physical qubit gate sets have a=.5,b=.75 versus a=.75,b=.5. Every allowed single-copy record agrees because only ab and known rotations enter. Yet labeled purities differ: $5/8$ and $25/32$. All product-only n-copy experiments remain gauge-equivalent. Independently trusted CNOT plus product measurements distinguishes them: same-sign probabilities $41/64$ and $19/32$. A claimed unique 'first witness' does not follow from the restricted data.

D3. Gravity law fails observational invariance. W9 uses tr(rho²), which is not fixed by the original single-copy observation contract. At gamma=1 the two operationally equivalent assignments predict a coupling ratio $\exp(-5/32) = 0.8553453273\ldots$, a 14.47% discrepancy in a supposedly physical weight. A purported physical property of an equivalence class cannot differ between equivalent representatives. W9 is ill-defined without cross-copy state calibration and an independent geometry probe.

D4. Elementary relations are stipulated. A four-site path and the same path with a formerly composite 0→3 operator declared elementary generate the same matrix algebra but give word distance three versus one. The selected spatial metric is not an intrinsic consequence of M4(C).

D5. Gauge-phase information may not survive measurement. Pure Kraus channel $T\rho T^\dagger$ forgets a global phase. Global simultaneous complex conjugation of all preparations, operations and effects leaves all internal outcome probabilities unchanged. An independent oriented coherent phase reference is necessary to operationally distinguish the two conjugate phase sectors.

D6. Irreversibility is not a causal arrow. Two irreversible bit reset channels reset-to-0 and reset-to-1. Compositions are order-sensitive, but both bit values are mutually reachable. Resetting a maximally mixed bit to pure reduces system entropy from log2 to zero. Thus noninvertible transitions do not imply causal antisymmetry or a monotonically increasing thermodynamic entropy.

D7. Geometry and physical units do not emerge. Causal order + volume theorems apply to suitable already-Lorentzian manifolds, not arbitrary finite partially ordered sets. A chain embeds as a timelike sequence in spacetime of many dimensions. The density scale and proper-time calibration remain imported.

D8. Event-count growth is not cosmological expansion. The number of causal events in an ever-larger causal diamond of ordinary flat Minkowski spacetime increases as the observed temporal extent grows, even though that spacetime is not expanding. Defining a(t) from raw accumulated event counts would falsely infer expansion in this control.

D9. Dynamics is not derived or normalized. W9/W10 do not supply Einstein constraints, proper Lorentzian degrees of freedom, matter/antimatter spectrum, a conserved stress tensor, a renormalized continuum limit or viable cosmological observables. A finite positive history kernel and an arbitrary action do not entail a physically consistent quantum field theory.

D10. Instrument and source-gain aliases. Different stationary and setting-dependent detectors can produce identical entire matched-coupling records. Independent crossed controls break the particular pair: at (d,c)=(1,2), pA=4/45 and pB=1/5, while the full-scale 10% Bayes threshold is 45 trials; at (2,1) it is 39. This is operational design, not evidence for a cosmic process ontology.

D11. Inverse limits can add fictional worlds. Compatible residues 2^k-1 describe -1 in the 2-adic completion but no nonnegative integer original model. F4 must stipulate an admissible completion and prove realized consistent global histories.

D12. Real physics is not matched. No independently derived 3+1 Lorentzian universe, experimentally calibrated clock, SU(3)×SU(2)×U(1) gauge content, Standard Model chiral-

ity/masses, Einstein equation, observed cosmic expansion or unique predictions beyond supplied models. F4 is not an empirical theory of our cosmos.

Disposition: W1–W10 and the cosmological narrative constitute an explicitly fictional counterfactual. D1–D12 dismantle its claimed physical *derivation* while preserving a set of conditional mathematical questions. A coherent subset may survive as a process-theory toy model, not as fundamental physics.

VI. Falsification targets, not retrospective excuses

- **Quotient audit:** Does every proposed 'physical' functional remain invariant under every operationally allowed similarity, antiunitary co-conjugation and change of detector/source representation? W9 fails before cross-copy calibration.
- **Copy-extension audit:** Add n independent copies but no cross-copy interactions. Predict no change in gauge-identifiability; then add one *independently marked* entangler and test the exact probabilities 41/64 vs 19/32.
- **Phase experiment:** Compare signed loop phases with a calibrated complex reference; repeat with the reference conjugated to test whether absolute chirality was supplied rather than derived.
- **Causal-geometry audit:** Produce a locally finite partial order with convergent manifold-like statistics, a consistent physical volume standard, a stable effective metric and Lorentz-covariant dynamics. Repeat for pathological non-manifoldlike posets.
- **Gravity test:** First prove a gauge-invariant, measurable feedback quantity and a dimensionally consistent update law. On physically calibrated test systems, compare against measured stress response rather than fitting gamma after outcomes.
- **Clock/expansion audit:** Distinguish true changes of generator from scale readout drift and observational window growth using an independent physical clock and bypassed scale standard.

VII. Publication and handoff

The conditional graph, detector, matrix, GST, and causal-structure facts belong to established or independently checked mathematical/operational research, with prior art. A specialist paper need not prove spacetime emergence, but must isolate a genuinely original methods theorem and a physical access contract. This F4 fictional cosmology is **not a scientific manuscript that should claim discovery**. The October 30/31 Wilson proof ledgers are separate.

Final wildcard handoff: Preserve the mathematical process-atlas idea as an optional problem statement. Retire the claim of a derived cosmology. Future sessions should resume strict scientific method without automatic fictional universes or stipulated physical bridges, unless directly requested.

VIII. Primary literature checked

- Lin et al., *On the freedom in representing quantum operations*, arXiv:1810.05631 (2018); and gate-set-tomography gauge discussion in *Nature Communications* 8, 14485 (2017).
- Chiribella, D'Ariano and Perinotti, *Informational derivation of quantum theory*, *Physical Review A* 84, 012311 (2011), DOI 10.1103/PhysRevA.84.012311.
- Oreshkov, Costa and Brukner, *Quantum correlations with no causal order*, *Nature Communications* 3, 1092 (2012), DOI 10.1038/ncomms2076.
- Rideout and Sorkin, *Classical sequential growth dynamics for causal sets*, *Physical Review D* 61, 024002 (1999).
- Surya, *The causal set approach to quantum gravity*, *Living Reviews in Relativity* 22, 5 (2019), DOI 10.1007/s41114-019-0023-1.
- Malament, *The class of continuous timelike curves determines the topology of spacetime*, *Journal of Mathematical Physics* 18 (1977); and Hawking–King–McCarthy (1976) causal/conformal geometry results.

- Zanardi, Lidar and Lloyd, *Quantum tensor product structures are observable induced*, *Physical Review Letters* 92, 060402 (2004).

This report is creative hypothesis generation and independent source-specific calculation, not evidence for any new fundamental physical law.